数理统计导论

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Part 1

概率论基础

CHAPTER 1

Probability and Distributions

Chapter 1 概述

本章建立概率论的基本框架，从集合论和概率公理出发，逐步过渡到随机变量、期望、方差等核心概念。

1. 1.1 Introduction

本节直观介绍概率模型的概念，在 1.3 节中将其形式化。许多研究的特点是，在本质上相同的条件下可以重复进行实验。例如，医学研究关注药物的效果；经济学家关注某些商品的价格变化；农学家研究肥料对粮食产量的影响。实验产生 outcome，但无法在实验前确定预测结果。

若实验可以重复进行，且所有可能 outcome 可以事先描述，则称为随机实验（random experiment），所有可能 outcome 的集合称为样本空间（sample space），记为 $\mathcal{C}$。

Example 1.1 (例 1.1.1). *抛掷硬币，outcome 为 T（反面）或 H（正面）。样本空间为 $\mathcal{C} = \{H, T\}$。*

Example 1.2 (例 1.1.2). *投掷一枚红骰子和一枚白骰子，outcome 为有序对（红骰子点数，白骰子点数）。样本空间包含 36 个有序对：$\mathcal{C} = \{(1,1), \ldots, (1,6), (2,1), \ldots, (6,6)\}$。*

我们用小写 Roman 字母表示 $\mathcal{C}$ 的元素，如 a, b, c。事件（event）是 $\mathcal{C}$ 的子集，用大写 Roman 字母表示，如 A, B, C。

重复实验 N 次，事件 A 发生的次数（频率）记为 f，则 f/N 称为相对频率（relative frequency）。当 N 增大时，相对频率趋于稳定，趋向某个数 p，这称为事件 A 的概率（probability）。

Example 1.3 (例 1.1.3). *在例 1.1.2 中，设 B 为点数之和为 7 的所有有序对：$B = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$。若投掷 $N = 400$ 次，$f = 60$ 次出现点数之和为 7，则相对频率 $f/N = 60/400 = 0.15$，因此 B 的概率约为 0.15。*

Remark (注 1.1.1). *概率的相对频率解释依赖于实验可重复的条件。另一种观点是将概率视为主观信念。无论采用哪种解释，1.3 节的概率数学性质都是一致的。*

统计学的目的是为随机实验提供数学模型，进而进行统计推断。概率论的理论基础之一是基于集合和集合函数的概念，这正是下一节的内容。

2. 1.2 Sets

集合是现代数学的基本概念。设 C 表示区间 $[0,1]$ 上的实数集合，则 $\frac{3}{4} \in C$。

若集合 C 是有限的或与正整数集等势，则称 C 为可数的（countable）。

2.1. 1.2.1 Review of Set Theory. 设 $\mathcal{C}$ 为样本空间，事件是 $\mathcal{C}$ 的子集。

Definition 1.1 (定义 1.2.1). *事件 A 的补（complement）A^c 是 $\mathcal{C}$ 中所有不属于 A 的元素构成的集合：$A^c = \{x \in \mathcal{C} : x \notin A\}$。*

空集（empty set）记为 ϕ。注意 $\mathcal{C}^c = \phi$，$\phi^c = \mathcal{C}$。

Definition 1.2 (定义 1.2.2). *若集合 A 的每个元素都是集合 B 的元素，则称 A 是 B 的子集，记作 $A \subset B$。若 $A \subset B$ 且 $B \subset A$，则 $A = B$。*

Definition 1.3 (定义 1.2.3). *设 A 和 B 为事件。A 和 B 的并（union）$A \cup B$ 是所有属于 A 或属于 B 的元素构成的集合。*

Definition 1.4 (定义 1.2.4). *设 A 和 B 为事件。A 和 B 的交（intersection）$A \cap B$ 是所有同时属于 A 和 B 的元素构成的集合。*

Definition 1.5 (定义 1.2.5). *若 $A \cap B = \phi$，则称 A 和 B 是不交的（disjoint）。*

若 $A_1, A_2, \ldots, A_n$ 为任意集合，定义

$$A_1 \cup A_2 \cup \cdots \cup A_n = \{x : x \in A_i \text{ for some } i = 1, 2, \ldots, n\} \tag{1.2.8}$$

$$A_1 \cap A_2 \cap \cdots \cap A_n = \{x : x \in A_i \text{ for all } i = 1, 2, \ldots, n\}. \tag{1.2.9}$$

简写为 $\cup_{i=1}^n A_i$ 和 $\cap_{i=1}^n A_i$。

可数并和可数交定义为

$$\cup_{n=1}^\infty A_n = \{x : x \in A_n \text{ for some } n\} \tag{1.2.10}$$

$$\cap_{n=1}^\infty A_n = \{x : x \in A_n \text{ for all } n\}. \tag{1.2.11}$$

德摩根定律（DeMorgan's Laws）：对任意集合 A 和 B，

$$(A \cap B)^c = A^c \cup B^c \tag{1.2.6}$$

$$(A \cup B)^c = A^c \cap B^c. \tag{1.2.7}$$

分配律：

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \tag{1.2.5}$$

Example 1.4 (例 1.2.3). *设 $\mathcal{C} = \{1, 2, 3, \ldots\}$，$A_n = \{1, 3, \ldots, 2n-1\}$，$B_n = \{n, n+1, \ldots\}$，则*

$$\cup_{n=1}^\infty A_n = \{1, 3, 5, \ldots\}; \quad \cap_{n=1}^\infty A_n = \{1\}$$

$$\cup_{n=1}^\infty B_n = \mathcal{C}; \quad \cap_{n=1}^\infty B_n = \phi. \tag{1.2.13}$$

单调序列：若 $\{A_n\}$ 非递减（$A_n \subset A_{n+1}$），则 $\lim_{n\to\infty} A_n = \cup_{n=1}^\infty A_n$。若 $\{A_n\}$ 非递增（$A_n \supset A_{n+1}$），则 $\lim_{n\to\infty} A_n = \cap_{n=1}^\infty A_n$。

2.2. 1.2.2 Set Functions. 集合函数（set function）是将集合映射到实数的函数。

Example 1.5 (例 1.2.5). 设 $\mathcal{C} = \mathbb{R}$，$Q(A)$ 为 A 中正整数的个数。若 $A = \{x : 0 < x < 5\}$，则 $Q(A) = 4$；若 $A = \{-2, -1\}$，则 $Q(A) = 0$。

Example 1.6 (例 1.2.6). 设 $\mathcal{C} = \mathbb{R}^2$，$Q(A)$ 为 A 的面积（若面积存在）。若 $A = \{(x, y) : x^2 + y^2 \leq 1\}$（单位圆盘），则 $Q(A) = \pi$。

集合函数常用求和或积分定义。记号:

$$\sum_A f(x) \quad \text{表示在所有} x \in A \text{ 上求和}$$

$$\int_A f(x)\, dx \quad \text{表示在集合} A \text{ 上的积分}$$

Example 1.7 (例 1.2.7). 设 $\mathcal{C}$ 为所有非负整数构成的集合，$Q(A) = \sum_{n \in A} \left(\frac{2}{3}\right)^n$。由几何级数公式 $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$（$|a| < 1$），可得 $Q(\mathcal{C}) = 3$。

Example 1.8 (例 1.2.8). 设 $\mathcal{C} = (0, \infty)$，$Q(A) = \int_A e^{-x}\, dx$。则

$$Q[(1, 3)] = \int_1^3 e^{-x}\, dx = e^{-1} - e^{-3} \approx 0.318$$

$$Q(\mathcal{C}) = \int_0^{\infty} e^{-x}\, dx = 1.$$

3. 1.3 The Probability Set Function

设 $\mathcal{C}$ 为样本空间，$\mathcal{B}$ 为事件域（σ-域），现在定义概率集函数。

Remark 1.3.1: 概率的定义由三条公理组成，这三条公理来源于相对频率的三条直观性质。设重复实验 N 次，事件 A 的相对频率为 $f_A = \#\{A\}/N$。则 $f_A \geq 0$、$f_{\mathcal{C}} = 1$，且若 A_1, A_2 不交则 $f_{A_1 \cup A_2} = f_{A_1} + f_{A_2}$。

Definition 1.6 (定义 1.3.1 (概率)). 设 $\mathcal{C}$ 为样本空间，$\mathcal{B}$ 为事件域。函数 $\mathbb{P} : \mathcal{B} \to \mathbb{R}$ 满足以下三条条件则称为概率集函数:

(1) $\mathbb{P}(A) \geq 0$，对所有 $A \in \mathcal{B}$

(2) $\mathbb{P}(\mathcal{C}) = 1$

(3) 若 $\{A_n\}$ 为两两不相交的事件序列，则

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

若事件序列两两不相交，称其为互不相容的（mutually exclusive）。若序列还是穷尽的（exhaustive），即其并为 $\mathcal{C}$，则构成 $\mathcal{C}$ 的一个划分（partition）。

概率集函数告诉我们在事件域上概率如何分布。

Theorem 1.1 (定理 1.3.1). 对每个事件 $A \in \mathcal{B}$，$\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$。

证明. 由 $\mathcal{C} = A \cup A^c$ 和 $A \cap A^c = \phi$，结合公理 (2) 和 (3) 得

$$1 = \mathbb{P}(A) + \mathbb{P}(A^c).$$

□

THEOREM 1.2 (定理 1.3.2). $\mathbb{P}(\phi) = 0$。

THEOREM 1.3 (定理 1.3.3). *若 $A \subset B$，则 $\mathbb{P}(A) \leq \mathbb{P}(B)$。*

证明. $B = A \cup (A^c \cap B)$，且两项不相交。由公理 (3) 得 $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B) \geq \mathbb{P}(A)$。 □

COROLLARY 1.1 (推论 1.3.1). *对每个 $A \in \mathcal{B}$，$0 \leq \mathbb{P}(A) \leq 1$。*

THEOREM 1.4 (定理 1.3.5). *对任意事件 A, B，*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

证明. 将 $A \cup B$ 和 B 分解为不相交集合的并：

$$A \cup B = A \cup (A^c \cap B), \quad B = (A \cap B) \cup (A^c \cap B).$$

由公理 (3) 得 $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B)$ 和 $\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(A^c \cap B)$。消去 $\mathbb{P}(A^c \cap B)$ 即得结果。 □

等可能性情形（Equilikely Case）：若样本空间 $\mathcal{C} = \{x_1, x_2, \ldots, x_m\}$ 有限，且每个 outcome 等概率 $p_i = 1/m$，则

$$\mathbb{P}(A) = \sum_{x_i \in A} \frac{1}{m} = \frac{\#(A)}{m}.$$

4. 1.4 Conditional Probability and Independence

4.1. Conditional Probability.

DEFINITION 1.7 (定义 1.4.1 (条件概率)). *设 $\mathbb{P}(B) > 0$，在事件 B 已发生的条件下，事件 A 的条件概率（conditional probability）定义为*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

条件概率也满足概率公理。

由条件概率定义可直接导出乘法法则（multiplication rule）：

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B \mid A) = \mathbb{P}(B) \cdot \mathbb{P}(A \mid B).$$

推广到 n 个事件：

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}\left(A_n \mid \bigcap_{i=1}^{n-1} A_i\right).$$

全概率公式（Law of Total Probability）：设 $\{B_i\}$ 为 $\mathcal{C}$ 的一个划分，则对任意事件 A，

$$\mathbb{P}(A) = \sum_i \mathbb{P}(B_i) \cdot \mathbb{P}(A \mid B_i).$$

THEOREM 1.5 (定理 1.4.1 (贝叶斯公式)). *设 $\{B_i\}$ 为 $\mathcal{C}$ 的一个划分，$\mathbb{P}(B_i) > 0$。则对任意事件 A ($\mathbb{P}(A) > 0$)，*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(B_j) \cdot \mathbb{P}(A \mid B_j)}{\sum_i \mathbb{P}(B_i) \cdot \mathbb{P}(A \mid B_i)}.$$

术语：$\mathbb{P}(B_j)$ 为先验概率（prior probability），$\mathbb{P}(B_j \mid A)$ 为后验概率（posterior probability），$\mathbb{P}(A \mid B_j)$ 为似然（likelihood）。

EXAMPLE 1.9 (例 1.4.1 (医学检测)). *设某疾病患病率为 1%，检测准确率为 99%。令 $D = \{$患病$\}$，$+ = \{$检测阳性$\}$。则*

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(D) \cdot \mathbb{P}(+ \mid D)}{\mathbb{P}(D) \cdot \mathbb{P}(+ \mid D) + \mathbb{P}(D^c) \cdot \mathbb{P}(+ \mid D^c)} = \frac{0.01 \cdot 0.99}{0.01 \cdot 0.99 + 0.99 \cdot 0.01} = \frac{1}{2}.$$

即使检测准确率高达 99%，检测阳性者真正患病的概率只有 50%!

4.2. 1.4.1 Independence.

DEFINITION 1.8 (定义 1.4.2 (独立性)). *若 $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$，则称事件 A 与 B 相互独立（independent），记作 $A \perp\!\!\!\perp B$。*

当 $\mathbb{P}(A) > 0, \mathbb{P}(B) > 0$ 时，独立性等价于 $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ 或 $\mathbb{P}(B \mid A) = \mathbb{P}(B)$。

PROPOSITION 1.1 (命题 1.4.1). *若 $A \perp\!\!\!\perp B$，则：*

(1) $A \perp\!\!\!\perp B^c$
(2) $A^c \perp\!\!\!\perp B$
(3) $A^c \perp\!\!\!\perp B^c$

DEFINITION 1.9 (定义 1.4.3 (相互独立)). *事件序列 $\{A_i : i \in I\}$ 相互独立，若对任意有限子集 $\{i_1, \ldots, i_k\}$，*

$$\mathbb{P}\left(\bigcap_{j=1}^{k} A_{i_j}\right) = \prod_{j=1}^{k} \mathbb{P}(A_{i_j}).$$

Warning: 两两独立（pairwise independent）不一定意味着相互独立!

5. 1.5 Random Variables

随机变量（random variable）是从样本空间 $\mathcal{C}$ 到实数的函数：$X : \mathcal{C} \to \mathbb{R}$。
通常用大写字母 X, Y, Z 表示随机变量，用小写字母 x, y, z 表示具体取值。
随机变量可分为：

- **离散随机变量（discrete）**：取值可数
- **连续随机变量（continuous）**：取值充满区间，不可数

Example 1.10 (例 1.5.1). 抛掷两枚硬币。样本空间 $\mathcal{C} = \{HH, HT, TH, TT\}$。定义 X 为正面出现的次数：$X(HH) = 2$，$X(HT) = X(TH) = 1$，$X(TT) = 0$。则 X 是取值在 $\{0, 1, 2\}$ 的离散随机变量。

6. 1.6 Discrete Random Variables

Definition 1.10 (定义 1.6.1 (概率质量函数)). 设 X 为离散随机变量。函数 $p_X(x) = \mathbb{P}(X = x)$ 称为概率质量函数（*probability mass function, pmf*）。

pmf 满足：$p_X(x) \geq 0$ 对所有 x，且 $\sum_x p_X(x) = 1$。

伯努利分布（Bernoulli）：$X \sim \mathrm{Bernoulli}(p)$，取值 $\{0, 1\}$，

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

二项分布（Binomial）：$X \sim \mathrm{Bin}(n, p)$，

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

这是 n 次独立伯努利试验中成功的次数。

泊松分布（Poisson）：$X \sim \mathrm{Poisson}(\lambda)$，

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots.$$

泊松分布是二项分布当 $n \to \infty$、$p \to 0$、$np = \lambda$ 固定时的极限分布。

6.1. 1.6.1 Transformations.

设 X 为离散随机变量，$Y = g(X)$。则 $p_Y(y) = \sum_{x:g(x)=y} p_X(x)$。

7. 1.7 Continuous Random Variables

Definition 1.11 (定义 1.7.1 (概率密度函数)). 设 X 为连续随机变量。若存在非负函数 $f_X(x)$ 使得

$$\mathbb{P}(a < X < b) = \int_a^b f_X(x)\, dx$$

对任意 $a < b$ 成立，则称 $f_X(x)$ 为概率密度函数（*probability density function, pdf*）。

pdf 满足：$f_X(x) \geq 0$ 对所有 x，且 $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$。

Warning: $f_X(x)$ 可以大于 1!

正态分布（Normal/Gaussian）：$X \sim N(\mu, \sigma^2)$，

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

标准正态分布：$Z \sim N(0, 1)$，$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$。

指数分布（Exponential）：$X \sim \mathrm{Exp}(\lambda)$，$f(x) = \lambda e^{-\lambda x}$，$x > 0$。

7.1. 1.7.1 Quantiles.

Definition 1.12 (定义 1.7.2 (分位数)). 设 X 的 *CDF* 为 $F_X(x)$。对 $0 < p < 1$，p-分位数 x_p 满足 $F_X(x_p) = p$。中位数是 $p = 0.5$ 的分位数。

7.2. 1.7.2 Transformations. 设 X 为连续随机变量，$Y = g(X)$。若 $y = g(x)$ 严格单调且可导，则

$$f_Y(y) = f_X(h(y)) \cdot |h'(y)|,$$

其中 $x = h(y)$ 是 $y = g(x)$ 的反函数。

若 $Y = aX + b$，则 $f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$。

8. 1.8 Expectation of a Random Variable

Definition 1.13 (定义 1.8.1 (数学期望)). 随机变量 X 的期望（*expected value*）:

- 离散型：$\mathbb{E}X = \sum_x x \cdot p_X(x)$
- 连续型：$\mathbb{E}X = \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx$

要求级数或积分绝对收敛。

Theorem 1.6 (定理 1.8.1 (期望的线性性)). 对任意随机变量 X, Y 和常数 a, b:

$$\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y.$$

关键点：无需 X, Y 相互独立！

Theorem 1.7 (定理 1.8.2 (方差公式)).

$$var(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2,$$

其中 $\mu = \mathbb{E}X$。

Theorem 1.8 (定理 1.8.3 (方差的性质)). *(1)* $var(aX + b) = a^2 var(X)$
(2) 若 $X_1, \ldots, X_n$ 相互独立，则 $var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n var(X_i)$

9. 1.9 Some Special Expectations

Definition 1.14 (定义 1.9.1 (协方差)).

$$cov(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}(XY) - \mathbb{E}X \cdot \mathbb{E}Y.$$

协方差的性质:

(1) $\mathrm{cov}(X, X) = \mathrm{var}(X)$
(2) $\mathrm{cov}(X, Y) = \mathrm{cov}(Y, X)$
(3) $\mathrm{cov}(aX + b, Y) = a\mathrm{cov}(X, Y)$
(4) $\mathrm{var}(X + Y) = \mathrm{var}(X) + \mathrm{var}(Y) + 2\mathrm{cov}(X, Y)$
(5) 若 $X \perp\!\!\!\perp Y$，则 $\mathrm{cov}(X, Y) = 0$

Definition 1.15 (定义 1.9.2 (相关系数)).

$$corr(X,Y) = \frac{cov(X,Y)}{\sqrt{var(X)\cdot var(Y)}}.$$

$-1 \le \mathrm{corr}(X,Y) \le 1$。

10. 1.10 Important Inequalities

Theorem 1.9 (Markov 不等式). 若 X 为随机变量，$g(X) \ge 0$，则对任意 $a > 0$，

$$\mathbb{P}(g(X) \ge a) \le \frac{\mathbb{E}[g(X)]}{a}.$$

特别地，取 $g(X) = X^2$，$a = k^2$，得

$$\mathbb{P}(|X| \ge k) \le \frac{\mathbb{E}[X^2]}{k^2}.$$

Theorem 1.10 (Chebyshev 不等式). 设 $\mu = \mathbb{E}X$，$\sigma^2 = var(X)$。则对任意 $k > 0$，

$$\mathbb{P}(|X-\mu| \ge k) \le \frac{\sigma^2}{k^2}.$$

Theorem 1.11 (Cauchy-Schwarz 不等式).

$$|\mathbb{E}(XY)| \le \sqrt{\mathbb{E}[X^2]\cdot\mathbb{E}[Y^2]}.$$

Summary

本章建立了概率论的基本框架：

(1) **概率公理化**：从相对频率的直观性质出发，建立了概率的公理化定义
(2) **条件概率与贝叶斯公式**：提供了更新信息的数学工具
(3) **独立性**：刻画了两个事件之间的概率关系
(4) **随机变量**：从集合论过渡到数值函数，为分布理论奠定基础
(5) **期望与方差**：分别度量分布的中心位置和离散程度
(6) **协方差与相关系数**：度量两个随机变量的联合变化趋势
(7) **重要不等式**：Markov、Chebyshev、Cauchy-Schwarz

Exercises

Exercise 1.1. *1.2.1* 求 $C_1 \cup C_2$ 和 $C_1 \cap C_2$：

(1) $C_1 = \{0,1,2\}$，$C_2 = \{2,3,4\}$
(2) $C_1 = \{x : 0 < x < 2\}$，$C_2 = \{x : 1 \le x < 3\}$

Exercise 1.2. *1.3.1* 设 $\mathbb{P}(A) = 0.3$，$\mathbb{P}(B) = 0.4$，$\mathbb{P}(A\cap B) = 0.1$。求：

(1) $\mathbb{P}(A \cup B)$
(2) $\mathbb{P}(A \mid B)$
(3) $\mathbb{P}(B \mid A)$

Exercise 1.3. *1.4.1* 某箱中有 *5* 个红球和 *3* 个白球。无放回地抽取 *3* 个球。求恰好抽到 *2* 个红球的概率。

Exercise 1.4. *1.5.1* 设 $X \sim N(100, 225)$。求:

(1) $\mathbb{P}(X < 95)$

(2) $\mathbb{P}(90 < X < 110)$

CHAPTER 2

Distributions of Two Random Variables

1. Distributions of Two Random Variables

We begin the discussion of a pair of random variables with the following example. A coin is tossed three times and our interest is in the ordered number pair (number of H's on first two tosses, number of H's on all three tosses), where H and T represent, respectively, heads and tails. Let $\mathcal{C} = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}$ denote the sample space. Let X_1 denote the number of H's on the first two tosses and X_2 denote the number of H's on all three flips. Then our interest can be represented by the pair of random variables (X_1, X_2) . For example, $(X_1(HTH), X_2(HTH))$ represents the outcome $(1, 2)$. Continuing in this way, X_1 and X_2 are real-valued functions defined on the sample space $\mathcal{C}$, which take us from the sample space to the space of ordered number pairs.

$$\mathcal{D} = \{(0,0), (0,1), (1,1), (1,2), (2,2), (2,3)\}.$$

Thus X_1 and X_2 are two random variables defined on the space $\mathcal{C}$, and, in this example, the space of these random variables is the two-dimensional set $\mathcal{D}$, which is a subset of two-dimensional Euclidean space R^2 . Hence (X_1, X_2) is a vector function from $\mathcal{C}$ to $\mathcal{D}$. We now formulate the definition of a random vector.

Definition 2.1 (Random Vector). *Given a random experiment with a sample space $\mathcal{C}$, consider two random variables X_1 and X_2 , which assign to each element c of $\mathcal{C}$ one and only one ordered pair of numbers $X_1(c) = x_1$, $X_2(c) = x_2$. Then we say that (X_1, X_2) is a random vector. The space of (X_1, X_2) is the set of ordered pairs $\mathcal{D} = \{(x_1, x_2) : x_1 = X_1(c), x_2 = X_2(c), c \in \mathcal{C}\}$.*

We often denote random vectors using vector notation $\mathbf{X} = (X_1, X_2)'$, where the prime denotes the transpose of the row vector (X_1, X_2) . Also, we often use (X, Y) to denote random vectors.

Let $\mathcal{D}$ be the space associated with the random vector (X_1, X_2) . Let A be a subset of $\mathcal{D}$. As in the case of one random variable, we speak of the event A . We wish to define the probability of the event A , which we denote by $P_{X_1,X_2}[A]$. As with random variables in Section 1.5 we can uniquely define P_{X_1,X_2} in terms of the cumulative distribution function (cdf), which is given by

$$F_{X_1,X_2}(x_1,x_2) = P[X_1 \le x_1, X_2 \le x_2], \tag{1}$$

for all $(x_1,x_2) \in R^2$. Because X_1 and X_2 are random variables, each of the events in the above intersection and the intersection of the events are events in the original sample space $\mathcal{C}$. Thus the expression is well defined.

For all real constants $a < b$, $c < d$,

$$P(a_1 < X_1 \le b_1, a_2 < X_2 \le b_2) = F(b_1,b_2) - F(a_1,b_2) - F(b_1,a_2) + F(a_1,a_2). \tag{2.1.2}$$

Hence, all induced probabilities of sets of the form $(a_1,b_1] \times (a_2,b_2]$ can be formulated in terms of the cdf. We often call this cdf the joint cumulative distribution function of (X_1,X_2) .

As with random variables, we are mainly concerned with two types of random vectors, namely discrete and continuous. We first discuss the discrete type.

A random vector (X_1,X_2) is a discrete random vector if its space $\mathcal{D}$ is finite or countable. Hence, X_1 and X_2 are both discrete also. The joint probability mass function (pmf) of (X_1,X_2) is defined by

$$p_{X_1,X_2}(x_1,x_2) = P[X_1 = x_1, X_2 = x_2], \tag{2}$$

for all $(x_1,x_2) \in \mathcal{D}$. As with random variables, the pmf uniquely defines the cdf. It also is characterized by the two properties

$$\text{(i) } 0 \le p_{X_1,X_2}(x_1,x_2) \le 1 \text{ and (ii)} \sum_{\mathcal{D}}\sum p_{X_1,X_2}(x_1,x_2) = 1. \tag{3}$$

For an event $B \in \mathcal{D}$, we have

$$P[(X_1,X_2) \in B] = \sum\sum_{B} p_{X_1,X_2}(x_1,x_2).$$

Example 2.1. *Consider the example at the beginning of this section where a fair coin is flipped three times and X_1 and X_2 are the number of heads on the first two flips and all 3 flips, respectively. We can conveniently table the pmf of (X_1,X_2) as follows:*

	$x_2=0$	$x_2=1$	$x_2=2$	$x_2=3$	$p_{X_1}(x_1)$
$x_1=0$	1/8	1/8	0	0	2/8
$x_1=1$	0	2/8	2/8	0	4/8
$x_1=2$	0	0	1/8	1/8	2/8
$p_{X_2}(x_2)$	1/8	3/8	3/8	1/8	

For instance, $P(X_1 \ge 2, X_2 \ge 2) = p(2,2) + p(2,3) = 2/8$.

At times it is convenient to speak of the support of a discrete random vector (X_1, X_2) . These are all the points (x_1, x_2) in the space of (X_1, X_2) such that $p(x_1, x_2) > 0$. In the last example the support consists of the six points $\{(0,0), (0,1), (1,1), (1,2), (2,2), (2,3)\}$.

We say a random vector (X_1, X_2) with space $\mathcal{D}$ is of the continuous type if its cdf $F_{X_1,X_2}(x_1, x_2)$ is continuous. For the most part, the continuous random vectors in this book have cdfs that can be represented as integrals of nonnegative functions. That is, $F_{X_1,X_2}(x_1, x_2)$ can be expressed as

$$F_{X_1,X_2}(x_1, x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f_{X_1,X_2}(w_1, w_2)\, dw_1\, dw_2, \tag{4}$$

for all $(x_1, x_2) \in R^2$. We call the integrand the joint probability density function (pdf) of (X_1, X_2) . Then

$$\frac{\partial^2 F_{X_1,X_2}(x_1, x_2)}{\partial x_1 \partial x_2} = f_{X_1,X_2}(x_1, x_2),$$

except possibly on events that have probability zero. A pdf is essentially characterized by the two properties

$$\text{(i) } f_{X_1,X_2}(x_1, x_2) \geq 0 \text{ and (ii) } \int\int_{\mathcal{D}} f_{X_1,X_2}(x_1, x_2)\, dx_1\, dx_2 = 1. \tag{5}$$

For an event $A \in \mathcal{D}$, we have

$$P[(X_1, X_2) \in A] = \int\int_A f_{X_1,X_2}(x_1, x_2)\, dx_1\, dx_2.$$

Note that $P[(X_1, X_2) \in A]$ is just the volume under the surface $z = f_{X_1,X_2}(x_1, x_2)$ over the set A .

Remark. *As with univariate random variables, we often drop the subscript* (X_1, X_2) *from joint cdfs, pdfs, and pmfs, when it is clear from the context. We also use notation such as* f_{12} *instead of* f_{X_1,X_2} . *Besides* (X_1, X_2) , *we often use* (X, Y) *to express random vectors.*

Example 2.2. *Consider a continuous random vector* (X, Y) *which is uniformly distributed over the unit circle in* R^2 . *Since the area of the unit circle is* π , *the joint pdf is*

$$f(x, y) = \begin{cases} \frac{1}{\pi} & -1 < y < 1, -\sqrt{1-y^2} < x < \sqrt{1-y^2} \\ 0 & \textit{elsewhere.} \end{cases}$$

Probabilities of certain events follow immediately from geometry. For instance, let A *be the interior of the circle with radius* $1/2$. *Then* $P[(X, Y) \in A] = \pi(1/2)^2/\pi = 1/4$

. Next, let B be the ring formed by the concentric circles with the respective radii of $1/2$ *and* $\sqrt{2}/2$ *. Then* $P[(X,Y) \in B] = \pi[(\sqrt{2}/2)^2 - (1/2)^2]/\pi = 1/4$ *.*

EXAMPLE 2.3. *Suppose an electrical component has two batteries. Let X and Y denote the lifetimes in standard units of the respective batteries. Assume that the pdf of (X,Y) is*

$$f(x,y) = \begin{cases} 4xye^{-(x^2+y^2)} & x > 0, y > 0 \\ 0 & \textit{elsewhere.} \end{cases}$$

The surface $z = f(x,y)$ peaks at $(x,y) = (\sqrt{2}/2, \sqrt{2}/2)$. For instance, the probability that both batteries survive beyond $\sqrt{2}/2$ units is given by

$$P\left(X > \frac{\sqrt{2}}{2}, Y > \frac{\sqrt{2}}{2}\right) = \int_{\sqrt{2}/2}^{\infty}\int_{\sqrt{2}/2}^{\infty} 4xye^{-(x^2+y^2)}\,dx\,dy = (e^{-1/2})^2 \approx 0.3679.$$

For a continuous random vector (X_1, X_2) , the support of (X_1, X_2) contains all points (x_1, x_2) for which $f(x_1, x_2) > 0$. We denote the support of a random vector by $\mathcal{S}$. As in the univariate case, $\mathcal{S} \subset \mathcal{D}$.

1.1. Marginal Distributions. Let (X_1, X_2) be a random vector. Then both X_1 and X_2 are random variables. We can obtain their distributions in terms of the joint distribution of (X_1, X_2) as follows. Recall that the event which defined the cdf of X_1 at x_1 is $\{X_1 \le x_1\}$. However,

$$\{X_1 \le x_1\} = \{X_1 \le x_1\} \cap \{-\infty < X_2 < \infty\} = \{X_1 \le x_1, -\infty < X_2 < \infty\}.$$

Taking probabilities, we have

$$F_{X_1}(x_1) = P[X_1 \le x_1, -\infty < X_2 < \infty]. \tag{6}$$

By Theorem 1.3.6 we can write this equation as $F_{X_1}(x_1) = \lim_{x_2 \uparrow \infty} F(x_1, x_2)$. Thus we have a relationship between the cdfs, which we can extend to either the pmf or pdf depending on whether (X_1, X_2) is discrete or continuous.

First consider the discrete case. Let $\mathcal{D}_{X_1}$ be the support of X_1 . For $x_1 \in \mathcal{D}_{X_1}$, Equation (6) is equivalent to

$$F_{X_1}(x_1) = \sum_{w_1 \le x_1} \left\{ \sum_{x_2} p_{X_1,X_2}(w_1, x_2) \right\}.$$

By the uniqueness of cdfs, the quantity in braces must be the pmf of X_1 evaluated at w_1 ; that is,

$$p_{X_1}(x_1) = \sum_{x_2} p_{X_1,X_2}(x_1, x_2), \tag{7}$$

for all $x_1 \in \mathcal{D}_{X_1}$. Hence, to find the probability that X_1 is x_1 , keep x_1 fixed and sum p_{X_1,X_2} over all of x_2 . In terms of a tabled joint pmf with rows comprised of X_1 support values and columns comprised of X_2 support values, this says that the distribution of X_1 can be obtained by the marginal sums of the rows. Likewise, the pmf of X_2 can be obtained by marginal sums of the columns.

EXAMPLE 2.4. *Consider a random experiment that consists of drawing at random one chip from a bowl containing 10 chips of the same shape and size. Each chip has an ordered pair of numbers on it: one with* $(1,1)$ *, one with* $(2,1)$ *, two with* $(3,1)$ *, one with* $(1,2)$ *, two with* $(2,2)$ *, and three with* $(3,2)$ *. Let the random variables* X_1 *and* X_2 *be defined as the respective first and second values of the ordered pair. The joint pmf* $p(x_1, x_2)$ *of* X_1 *and* X_2 *can be given by the following table:*

	$x_2 = 1$	$x_2 = 2$	$p_{X_1}(x_1)$
$x_1 = 1$	1/10	1/10	2/10
$x_1 = 2$	1/10	2/10	3/10
$x_1 = 3$	2/10	3/10	5/10
$p_{X_2}(x_2)$	4/10	6/10	

We next consider the continuous case. Let $\mathcal{D}_{X_1}$ be the support of X_1 . For $x_1 \in \mathcal{D}_{X_1}$, Equation (6) is equivalent to

$$F_{X_1}(x_1) = \int_{-\infty}^{x_1} \left\{ \int_{-\infty}^{\infty} f_{X_1,X_2}(w_1, x_2)\, dx_2 \right\} dw_1.$$

By the uniqueness of cdfs, the quantity in braces must be the pdf of X_1 evaluated at w_1 ; that is,

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2)\, dx_2, \tag{8}$$

for all $x_1 \in \mathcal{D}_{X_1}$. Hence, in the continuous case the marginal pdf of X_1 is found by integrating out x_2 . Similarly, the marginal pdf of X_2 is found by integrating out x_1 .

EXAMPLE 2.5 (Example 2.2 continued). *Consider the vector of continuous random variables* (X, Y) *discussed in Example 2.2. To find the marginal distribution of* X *, fix* x *between* -1 *and* 1 *and then integrate out* y *from* $-\sqrt{1-x^2}$ *to* $\sqrt{1-x^2}$ *. Hence, the marginal pdf of* X *is*

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi}\, dy = \frac{2}{\pi}\sqrt{1-x^2}, \quad -1 < x < 1.$$

Although (X, Y) *has a joint uniform distribution, the distribution of* X *is unimodal with peak at 0. Because the joint pdf is symmetric in* x *and* y *, the marginal pdf of* Y *is the same as that of* X *.*

EXAMPLE 2.6. *Let* X_1 *and* X_2 *have the joint pdf*

$$f(x_1, x_2) = \begin{cases} x_1 + x_2 & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \textit{elsewhere.} \end{cases}$$

Notice the space of the random vector is the interior of the square with vertices $(0,0), (1,0), (1,1)$ *and* $(0,1)$ *. The marginal pdf of* X_1 *is*

$$f_1(x_1) = \int_0^1 (x_1 + x_2)\, dx_2 = x_1 + \frac{1}{2}, \quad 0 < x_1 < 1,$$

zero elsewhere, and the marginal pdf of X_2 *is*

$$f_2(x_2) = \int_0^1 (x_1 + x_2)\, dx_1 = \frac{1}{2} + x_2, \quad 0 < x_2 < 1,$$

zero elsewhere.

Suppose we want to find the probability $P(X_1 + X_2 \leq 1)$ *. The region of integration is the interior of the triangle with vertices* $(0,0)$ *,* $(1,0)$ *and* $(0,1)$ *. Fixing* x_1 *and integrating with respect to* x_2 *, we have*

$$P(X_1 + X_2 \leq 1) = \int_0^1 \left[\int_0^{1-x_1} (x_1 + x_2)\, dx_2 \right] dx_1 = \frac{1}{3}.$$

EXAMPLE 2.7 (Example 2.3 continued). *Recall that the random variables* X *and* Y *of Example 2.3 were the lifetimes of two batteries. Using the change-of-variable* $w = y^2$ *, the marginal pdf of* X *is*

$$f_X(x) = \int_0^\infty 4xye^{-(x^2+y^2)}\, dy = 2xe^{-x^2} \int_0^\infty e^{-w}\, dw = 2xe^{-x^2},$$

for $x > 0$ *. By the symmetry of* x *and* y *in the model, the pdf of* Y *is the same as that of* X *. To determine the median lifetime,* θ *, of these batteries, we need to solve*

$$\frac{1}{2} = \int_0^\theta 2xe^{-x^2}\, dx = 1 - e^{-\theta^2}.$$

Solving this equation, we obtain $\theta = \sqrt{\log 2} \approx 0.8326$ *.*

1.2. Expectation. The concept of expectation extends in a straightforward manner. Let (X_1, X_2) be a random vector and let $Y = g(X_1, X_2)$ for some real-valued function; i.e., $g : R^2 \to R$. Then Y is a random variable and we could determine its expectation by obtaining the distribution of Y . But Theorem 1.8.1 is true for random vectors also.

Suppose (X_1, X_2) is of the continuous type. Then $E(Y)$ exists if

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |g(x_1, x_2)| f_{X_1,X_2}(x_1, x_2)\, dx_1\, dx_2 < \infty.$$

Then

$$E(Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x_1, x_2) f_{X_1,X_2}(x_1, x_2)\, dx_1\, dx_2. \tag{9}$$

Likewise if (X_1, X_2) is discrete, then $E(Y)$ exists if

$$\sum_{x_1}\sum_{x_2} |g(x_1, x_2)| p_{X_1,X_2}(x_1, x_2) < \infty.$$

Then

$$E(Y) = \sum_{x_1}\sum_{x_2} g(x_1, x_2) p_{X_1,X_2}(x_1, x_2). \tag{10}$$

We can now show that E is a linear operator.

THEOREM 2.1. *Let (X_1, X_2) be a random vector. Let $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ be random variables whose expectations exist. Then for all real numbers k_1 and k_2 ,*

$$E(k_1Y_1 + k_2Y_2) = k_1E(Y_1) + k_2E(Y_2). \tag{11}$$

EXAMPLE 2.8. *Let X_1 and X_2 have the pdf*

$$f(x_1, x_2) = \begin{cases} 8x_1x_2 & 0 < x_1 < x_2 < 1 \\ 0 & \textit{elsewhere.} \end{cases}$$

Then

$$E(X_1X_2^2) = \int_0^1 \left[\int_0^{x_2} 8x_1^2x_2^3\, dx_1\right] dx_2 = \int_0^1 \frac{8}{3}x_2^6\, dx_2 = \frac{8}{21}.$$

In addition,

$$E(X_2) = \int_0^1 \left[\int_0^{x_2} x_2(8x_1x_2)\, dx_1\right] dx_2 = \frac{4}{5}.$$

Since X_2 has the pdf $f_2(x_2) = 4x_2^3$, $0 < x_2 < 1$, zero elsewhere, the latter expectation can also be found by

$$E(X_2) = \int_0^1 x_2(4x_2^3)\,dx_2 = \frac{4}{5}.$$

Using Theorem 2.1,

$$E(7X_1X_2^2 + 5X_2) = 7E(X_1X_2^2) + 5E(X_2) = 7 \cdot \frac{8}{21} + 5 \cdot \frac{4}{5} = \frac{20}{3}.$$

We also note that the expected value of any function $g(X_2)$ of X_2 can be found in two ways:

$$E(g(X_2)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x_2)f(x_1,x_2)\,dx_1\,dx_2 = \int_{-\infty}^{\infty} g(x_2)f_{X_2}(x_2)\,dx_2,$$

the latter single integral being obtained from the double integral by integrating on x_1 first.

Definition 2.2 (Moment Generating Function of a Random Vector). *Let* $\mathbf{X} = (X_1, X_2)'$ *be a random vector. If* $E(e^{t_1X_1+t_2X_2})$ *exists for* $|t_1| < h_1$ *and* $|t_2| < h_2$ *, where* h_1 *and* h_2 *are positive, it is denoted by* $M_{X_1,X_2}(t_1,t_2)$ *and is called the moment generating function (mgf) of* $\mathbf{X}$ *.*

As in the one-variable case, if it exists, the mgf of a random vector uniquely determines the distribution of the random vector.

Let $\mathbf{t} = (t_1, t_2)'$. Then we can write the mgf of $\mathbf{X}$ as

$$M_{X_1,X_2}(\mathbf{t}) = E\left[e^{\mathbf{t}'\mathbf{X}}\right]. \tag{12}$$

Also, the mgfs of X_1 and X_2 are immediately seen to be $M_{X_1,X_2}(t_1,0)$ and $M_{X_1,X_2}(0,t_2)$, respectively.

Example 2.9. *Let the continuous-type random variables* X *and* Y *have the joint pdf*

$$f(x,y) = \begin{cases} e^{-y} & 0 < x < y < \infty \\ 0 & \textit{elsewhere.} \end{cases}$$

The mgf of this joint distribution is

$$M(t_1,t_2) = \int_0^{\infty}\left[\int_x^{\infty} \exp(t_1x + t_2y - y)\,dy\right]dx = \frac{1}{(1-t_1-t_2)(1-t_2)},$$

provided that $t_1 + t_2 < 1$ *and* $t_2 < 1$ *. Furthermore, the moment-generating functions of the marginal distributions of* X *and* Y *are, respectively,*

$$M(t_1,0) = \frac{1}{1-t_1}, \quad t_1 < 1,$$
$$M(0,t_2) = \frac{1}{(1-t_2)^2}, \quad t_2 < 1.$$

These are respectively those of the marginal probability density functions

$$f_1(x) = \int_x^\infty e^{-y}\,dy = e^{-x}, \quad 0 < x < \infty,$$

zero elsewhere, and

$$f_2(y) = e^{-y}\int_0^y dx = ye^{-y}, \quad 0 < y < \infty,$$

zero elsewhere.

Definition 2.3 (Expected Value of a Random Vector). *Let* $\mathbf{X} = (X_1, X_2)'$ *be a random vector. Then the expected value of* $\mathbf{X}$ *exists if the expectations of* X_1 *and* X_2 *exist. If it exists, then the expected value is given by*

$$E[\mathbf{X}] = \begin{bmatrix} E(X_1) \\ E(X_2) \end{bmatrix}. \tag{13}$$

2. Transformations

Let (X_1, X_2) be a random vector with known distribution. Often we are interested in a function of (X_1, X_2) such as $Y = u(X_1, X_2)$. In this section we discuss the transformation technique for finding the distribution of Y . We begin with a discussion of one-to-one transformations of random vectors.

Suppose (X_1, X_2) is a continuous type random vector with pdf $f_{X_1,X_2}(x_1, x_2)$ defined on the support $\mathcal{S}$. Let $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ define a transformation that maps $\mathcal{S}$ onto $\mathcal{T}$. If this transformation is one-to-one, then there exists an inverse transformation $X_1 = v_1(Y_1, Y_2)$ and $X_2 = v_2(Y_1, Y_2)$ that maps $\mathcal{T}$ onto $\mathcal{S}$. The joint pdf of (Y_1, Y_2) is given by

$$f_{Y_1,Y_2}(y_1, y_2) = f_{X_1,X_2}(v_1(y_1, y_2), v_2(y_1, y_2)) \cdot |J|, \tag{14}$$

where J is the Jacobian of the transformation:

$$J = \det\begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{pmatrix} = \frac{\partial x_1}{\partial y_1}\cdot\frac{\partial x_2}{\partial y_2} - \frac{\partial x_1}{\partial y_2}\cdot\frac{\partial x_2}{\partial y_1}.$$

Example 2.10. *Let* X_1 *and* X_2 *have the joint pdf*

$$f_{X_1,X_2}(x_1, x_2) = \begin{cases} \frac{1}{8} & -2 < x_1 < 2, -1 < x_2 < 1 \\ 0 & \textit{elsewhere.} \end{cases}$$

Suppose we want to find the joint pdf of $Y_1 = X_1 + X_2$ *and* $Y_2 = X_1 - X_2$. *The inverse transformation is* $x_1 = (y_1 + y_2)/2$ *and* $x_2 = (y_1 - y_2)/2$. *The Jacobian is*

$$J = \det \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} = -\frac{1}{2} - \frac{1}{2} = -1.$$

The support $\mathcal{T}$ *is determined by* $|y_1 + y_2| < 4$ *and* $|y_1 - y_2| < 2$ *. Hence, the joint pdf of* (Y_1, Y_2) *is*

$$f_{Y_1,Y_2}(y_1, y_2) = \frac{1}{8} \cdot |J| = \frac{1}{8},$$

over the region $\mathcal{T}$ *.*

Example 2.11. *Let* X_1 *and* X_2 *have the joint pdf*

$$f_{X_1,X_2}(x_1, x_2) = \begin{cases} 8x_1x_2 & 0 < x_1 < x_2 < 1 \\ 0 & \textit{elsewhere.} \end{cases}$$

Suppose $Y_1 = X_1/X_2$ *and* $Y_2 = X_2$ *. The inverse transformation is* $x_1 = y_1y_2$ *and* $x_2 = y_2$ *. The Jacobian is* $J = y_2$ *. The support of* (Y_1, Y_2) *is* $\mathcal{T} = \{(y_1, y_2) : 0 < y_1 < 1, 0 < y_2 < 1\}$ *. Hence, the joint pdf is*

$$f_{Y_1,Y_2}(y_1, y_2) = 8y_1y_2 \cdot y_2 = 8y_1y_2^3,$$

over $\mathcal{T}$ *. The marginal pdfs are*

$$f_{Y_1}(y_1) = \int_0^1 8y_1y_2^3\, dy_2 = 2y_1, \quad 0 < y_1 < 1,$$

and

$$f_{Y_2}(y_2) = \int_0^1 8y_1y_2^3\, dy_1 = 4y_2^3, \quad 0 < y_2 < 1.$$

In addition to the change-of-variable and cdf techniques for finding distributions of functions of random variables, there is another method, called the moment generating function (mgf) technique, which works well for linear functions of random variables.

Example 2.12. *Let* X_1 *and* X_2 *have the joint pmf*

$$p_{X_1,X_2}(x_1, x_2) = \begin{cases} \frac{\mu_1^{x_1}\mu_2^{x_2}e^{-\mu_1}e^{-\mu_2}}{x_1!\,x_2!} & x_1 = 0, 1, 2, \ldots, x_2 = 0, 1, 2, \ldots \\ 0 & \textit{elsewhere.} \end{cases}$$

Let $Y = X_1 + X_2$ *and consider*

$$E(e^{tY}) = \sum_{x_1=0}^{\infty} \sum_{x_2=0}^{\infty} e^{t(x_1+x_2)} p_{X_1,X_2}(x_1, x_2) = e^{(\mu_1+\mu_2)(e^t-1)}.$$

By the uniqueness of mgfs, the pmf of Y *must be*

$$p_Y(y) = e^{-(\mu_1+\mu_2)}\frac{(\mu_1+\mu_2)^y}{y!}, \quad y = 0, 1, 2, \ldots,$$

which is a Poisson distribution with parameter $\mu_1 + \mu_2$.

3. Conditional Distributions and Expectations

In Section 1 we introduced the joint probability distribution of a pair of random variables. We also showed how to recover the individual (marginal) distributions for the random variables from the joint distribution. In this section, we discuss conditional distributions, i.e., the distribution of one of the random variables when the other has assumed a specific value.

Let X_1 and X_2 denote random variables of the discrete type, which have the joint pmf $p_{X_1,X_2}(x_1, x_2)$ that is positive on the support set $\mathcal{S}$ and is zero elsewhere. Let $p_{X_1}(x_1)$ and $p_{X_2}(x_2)$ denote, respectively, the marginal probability mass functions of X_1 and X_2 . Let x_1 be a point in the support of X_1 ; hence, $p_{X_1}(x_1) > 0$. Using the definition of conditional probability, we have

$$P(X_2 = x_2 \mid X_1 = x_1) = \frac{P(X_1 = x_1, X_2 = x_2)}{P(X_1 = x_1)} = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_1}(x_1)}.$$

Define this function as

$$p_{X_2|X_1}(x_2 \mid x_1) = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_1}(x_1)}, \quad x_2 \in \mathcal{S}_{X_2}. \tag{15}$$

For any fixed x_1 with $p_{X_1}(x_1) > 0$, this function $p_{X_2|X_1}(x_2 \mid x_1)$ satisfies the conditions of being a pmf of the discrete type because $p_{X_2|X_1}(x_2 \mid x_1)$ is nonnegative and

$$\sum_{x_2} p_{X_2|X_1}(x_2 \mid x_1) = \frac{1}{p_{X_1}(x_1)} \sum_{x_2} p_{X_1,X_2}(x_1, x_2) = \frac{p_{X_1}(x_1)}{p_{X_1}(x_1)} = 1.$$

We call $p_{X_2|X_1}(x_2 \mid x_1)$ the conditional pmf of the discrete type of random variable X_2 , given that the discrete type of random variable $X_1 = x_1$.

Now let X_1 and X_2 denote random variables of the continuous type and have the joint pdf $f_{X_1,X_2}(x_1, x_2)$ and the marginal probability density functions $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$, respectively. When $f_{X_1}(x_1) > 0$, we define the symbol $f_{X_2|X_1}(x_2 \mid x_1)$ by the relation

$$f_{X_2|X_1}(x_2 \mid x_1) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_1}(x_1)}. \tag{16}$$

In this relation, x_1 is to be thought of as having a fixed (but any fixed) value for which $f_{X_1}(x_1) > 0$. It is evident that $f_{X_2|X_1}(x_2 \mid x_1)$ is nonnegative and that

$$\int_{-\infty}^{\infty} f_{X_2|X_1}(x_2 \mid x_1)\, dx_2 = \frac{1}{f_{X_1}(x_1)} \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2)\, dx_2 = \frac{f_{X_1}(x_1)}{f_{X_1}(x_1)} = 1.$$

That is, $f_{X_2|X_1}(x_2 \mid x_1)$ has the properties of a pdf of one continuous type of random variable. It is called the conditional pdf of the continuous type of random variable X_2 , given that the continuous type of random variable X_1 has the value x_1 .

Since each of $f_{X_2|X_1}(x_2 \mid x_1)$ and $f_{X_1|X_2}(x_1 \mid x_2)$ is a pdf of one random variable, each has all the properties of such a pdf. If $u(X_2)$ is a function of X_2 , the conditional expectation of $u(X_2)$, given that $X_1 = x_1$, if it exists, is given by

$$E[u(X_2) \mid x_1] = \int_{-\infty}^{\infty} u(x_2) f_{X_2|X_1}(x_2 \mid x_1)\, dx_2. \tag{17}$$

Note that $E[u(X_2) \mid x_1]$ is a function of x_1 . If they do exist, then $E(X_2 \mid x_1)$ is the mean and $\mathrm{Var}(X_2 \mid x_1)$ is the conditional variance of the conditional distribution of X_2 , given $X_1 = x_1$. Of course, we have

$$\mathrm{Var}(X_2 \mid x_1) = E(X_2^2 \mid x_1) - [E(X_2 \mid x_1)]^2.$$

Example 2.13. *Let X_1 and X_2 have the joint pdf*

$$f(x_1, x_2) = \begin{cases} 2 & 0 < x_1 < x_2 < 1 \\ 0 & \textit{elsewhere.} \end{cases}$$

Then the marginal probability density functions are, respectively,

$$f_1(x_1) = \begin{cases} \int_{x_1}^{1} 2\, dx_2 = 2(1 - x_1) & 0 < x_1 < 1 \\ 0 & \textit{elsewhere,} \end{cases}$$

and

$$f_2(x_2) = \begin{cases} \int_{0}^{x_2} 2\, dx_1 = 2x_2 & 0 < x_2 < 1 \\ 0 & \textit{elsewhere.} \end{cases}$$

The conditional pdf of X_1 , given $X_2 = x_2$, $0 < x_2 < 1$, is

$$f_{1|2}(x_1 \mid x_2) = \frac{2}{2x_2} = \frac{1}{x_2}, \quad 0 < x_1 < x_2 < 1.$$

Here the conditional mean and the conditional variance of X_1 , given $X_2 = x_2$, are respectively,

$$E(X_1 \mid x_2) = \int_0^{x_2} x_1 \cdot \frac{1}{x_2}\, dx_1 = \frac{x_2}{2}, \quad 0 < x_2 < 1,$$

and

$$\mathrm{Var}(X_1 \mid x_2) = \int_0^{x_2} \left(x_1 - \frac{x_2}{2}\right)^2 \cdot \frac{1}{x_2}\, dx_1 = \frac{x_2^2}{12}, \quad 0 < x_2 < 1.$$

THEOREM 2.2. *Let (X_1, X_2) be a random vector such that the variance of X_2 is finite. Then:*

(1) $E[E(X_2 \mid X_1)] = E(X_2)$.
(2) $\mathrm{Var}[E(X_2 \mid X_1)] \leq \mathrm{Var}(X_2)$.

REMARK. *Intuitively, this result could have this useful interpretation. Both the random variables X_2 and $E(X_2 \mid X_1)$ have the same mean μ_2 . If we did not know μ_2 , we could use either of the two random variables to guess at the unknown μ_2 . Since, however,* $\mathrm{Var}(X_2) \geq \mathrm{Var}[E(X_2 \mid X_1)]$ *, we would put more reliance on $E(X_2 \mid X_1)$ as a guess. That is, if we observe the pair (X_1, X_2) to be (x_1, x_2) , we could prefer to use $E(X_2 \mid x_1)$ to x_2 as a guess at the unknown μ_2 . When studying the use of sufficient statistics in estimation in Chapter 7, we make use of this famous result, attributed to C. R. Rao and David Blackwell.*

4. Independent Random Variables

Let X_1 and X_2 denote the random variables of the continuous type that have the joint pdf $f(x_1, x_2)$ and marginal probability density functions $f_1(x_1)$ and $f_2(x_2)$, respectively. In accordance with the definition of the conditional pdf $f_{X_2|X_1}(x_2 \mid x_1)$, we may write the joint pdf $f(x_1, x_2)$ as

$$f(x_1, x_2) = f_{X_2|X_1}(x_2 \mid x_1) f_1(x_1).$$

Suppose that we have an instance where $f_{X_2|X_1}(x_2 \mid x_1)$ does not depend upon x_1 . Then the marginal pdf of X_2 is, for random variables of the continuous type,

$$f_2(x_2) = \int_{-\infty}^{\infty} f_{X_2|X_1}(x_2 \mid x_1) f_1(x_1)\, dx_1 = f_{X_2|X_1}(x_2 \mid x_1) \int_{-\infty}^{\infty} f_1(x_1)\, dx_1 = f_{X_2|X_1}(x_2 \mid x_1).$$

Accordingly,

$$f_2(x_2) = f_{X_2|X_1}(x_2 \mid x_1) \quad \text{and} \quad f(x_1, x_2) = f_1(x_1) f_2(x_2),$$

when $f_{X_2|X_1}(x_2 \mid x_1)$ does not depend upon x_1 . That is, if the conditional distribution of X_2 , given $X_1 = x_1$, is independent of any assumption about x_1 , then $f(x_1, x_2) = f_1(x_1) f_2(x_2)$.

DEFINITION 2.4 (Independence). *Let the random variables X_1 and X_2 have the joint pdf $f(x_1, x_2)$ [joint pmf $p(x_1, x_2)$] and the marginal pdfs [pmfs] $f_1(x_1)$ $[p_1(x_1)]$ and $f_2(x_2)$ $[p_2(x_2)]$, respectively. The random variables X_1 and X_2 are said to be independent if, and only if, $f(x_1, x_2) \equiv f_1(x_1) f_2(x_2)$ $[p(x_1, x_2) \equiv p_1(x_1) p_2(x_2)]$. Random variables that are not independent are said to be dependent.*

REMARK. *Two comments should be made about the preceding definition. First, the product of two positive functions* $f_1(x_1)f_2(x_2)$ *means a function that is positive on the product space. That is, if* $f_1(x_1)$ *and* $f_2(x_2)$ *are positive on, and only on, the respective spaces* $\mathcal{S}_1$ *and* $\mathcal{S}_2$ *, then the product of* $f_1(x_1)$ *and* $f_2(x_2)$ *is positive on, and only on, the product space* $\mathcal{S} = \{(x_1, x_2) : x_1 \in \mathcal{S}_1, x_2 \in \mathcal{S}_2\}$ *.*

EXAMPLE 2.14. *Let the joint pdf of* X_1 *and* X_2 *be*

$$f(x_1, x_2) = \begin{cases} x_1 + x_2 & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \textit{elsewhere.} \end{cases}$$

We show that X_1 *and* X_2 *are dependent. Here the marginal probability density functions are*

$$f_1(x_1) = \begin{cases} \int_0^1 (x_1 + x_2)\, dx_2 = x_1 + \frac{1}{2} & 0 < x_1 < 1 \\ 0 & \textit{elsewhere,} \end{cases}$$

and

$$f_2(x_2) = \begin{cases} \int_0^1 (x_1 + x_2)\, dx_1 = \frac{1}{2} + x_2 & 0 < x_2 < 1 \\ 0 & \textit{elsewhere.} \end{cases}$$

Since $f(x_1, x_2) \neq f_1(x_1)f_2(x_2)$ *, the random variables* X_1 *and* X_2 *are dependent.*

THEOREM 2.3. *Let the random variables* X_1 *and* X_2 *have supports* $\mathcal{S}_1$ *and* $\mathcal{S}_2$ *, respectively, and have the joint pdf* $f(x_1, x_2)$ *. Then* X_1 *and* X_2 *are independent if and only if* $f(x_1, x_2)$ *can be written as a product of a nonnegative function of* x_1 *and a nonnegative function of* x_2 *. That is,*

$$f(x_1, x_2) \equiv g(x_1)h(x_2),$$

where $g(x_1) > 0$ *,* $x_1 \in \mathcal{S}_1$ *, zero elsewhere, and* $h(x_2) > 0$ *,* $x_2 \in \mathcal{S}_2$ *, zero elsewhere.*

The following theorem provides a useful criterion for establishing independence.

THEOREM 2.4. *Let the random variables* X_1 *and* X_2 *have the joint cdf* $F(x_1, x_2)$ *and marginal cdfs* $F_1(x_1)$ *and* $F_2(x_2)$ *, respectively. Then* X_1 *and* X_2 *are independent if and only if*

$$F(x_1, x_2) \equiv F_1(x_1)F_2(x_2).$$

In the discrete case, this is equivalent to $p(x_1, x_2) \equiv p_1(x_1)p_2(x_2)$ *. In the continuous case, this is equivalent to* $f(x_1, x_2) \equiv f_1(x_1)f_2(x_2)$ *.*

If X_1 and X_2 are independent, then the conditional distributions equal the marginal distributions:

$$f_{X_2|X_1}(x_2 \mid x_1) = f_2(x_2) \quad \text{and} \quad f_{X_1|X_2}(x_1 \mid x_2) = f_1(x_1).$$

Moreover, for any functions $g(X_1)$ and $h(X_2)$,

$$E[g(X_1)h(X_2)] = E[g(X_1)] \cdot E[h(X_2)].$$

In particular, if X_1 and X_2 are independent, then

$$\mathrm{Cov}(X_1, X_2) = 0.$$

However, the converse is not true in general; zero covariance does not imply independence (unless the distribution is bivariate normal).

5. Correlation Coefficient

In Section 1 we defined the expected value of a random vector. Now we discuss measures of linear association between two random variables.

Definition 2.5 (Covariance). *The covariance between X_1 and X_2 is defined as*

$$\mathrm{Cov}(X_1, X_2) = E[(X_1 - E[X_1])(X_2 - E[X_2])] = E(X_1X_2) - E(X_1)E(X_2).$$

The covariance has the following properties:

(1) $\mathrm{Cov}(X_1, X_1) = \mathrm{Var}(X_1)$.
(2) $\mathrm{Cov}(X_1, X_2) = \mathrm{Cov}(X_2, X_1)$.
(3) $\mathrm{Cov}(aX_1, X_2) = a\,\mathrm{Cov}(X_1, X_2)$.
(4) $\mathrm{Cov}(X_1 + X_2, Y) = \mathrm{Cov}(X_1, Y) + \mathrm{Cov}(X_2, Y)$.
(5) If X_1 and X_2 are independent, then $\mathrm{Cov}(X_1, X_2) = 0$.

From property (4), we have

$$\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + 2\,\mathrm{Cov}(X_1, X_2).$$

Definition 2.6 (Correlation Coefficient). *The correlation coefficient between X_1 and X_2 is defined as*

$$\rho_{12} = corr(X_1, X_2) = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)}\sqrt{\mathrm{Var}(X_2)}} = \frac{\mathrm{Cov}(X_1, X_2)}{\sigma_1\sigma_2},$$

provided $\sigma_1 > 0$ and $\sigma_2 > 0$.

The correlation coefficient satisfies $-1 \leq \rho_{12} \leq 1$. Moreover, $|\rho_{12}| = 1$ if and only if X_1 and X_2 are linearly related; that is, there exist constants $a \neq 0$ and b such that $X_2 = aX_1 + b$ with probability 1. When $\rho_{12} = 0$, we say that X_1 and X_2 are uncorrelated.

Remark. *Note that covariance depends on the units of measurement, while the correlation coefficient is dimensionless. The correlation coefficient is a standardized measure of linear association.*

If X_1 and X_2 are independent, then they are uncorrelated ($\rho_{12} = 0$). However, the converse is not generally true; uncorrelated random variables need not be independent.

6. The Bivariate Normal Distribution

One of the most important joint distributions is the bivariate normal distribution.

Definition 2.7 (Bivariate Normal Distribution). *We say that (X_1, X_2) has a bivariate normal distribution if the joint pdf is given by*

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right\},$$

for $-\infty < x_1 < \infty$, $-\infty < x_2 < \infty$, where $\mu_1 \in R$, $\mu_2 \in R$, $\sigma_1 > 0$, $\sigma_2 > 0$, and $-1 < \rho < 1$. We write $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$.

The parameters have the following interpretations:

- $E(X_1) = \mu_1$ and $E(X_2) = \mu_2$.
- $\mathrm{Var}(X_1) = \sigma_1^2$ and $\mathrm{Var}(X_2) = \sigma_2^2$.
- ρ is the correlation coefficient between X_1 and X_2 .

Theorem 2.5 (Properties of Bivariate Normal). *If $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, then:*

(1) The marginal distributions are $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$.
(2) The conditional distribution of X_2 given $X_1 = x_1$ is normal with mean and variance:

$$E(X_2 \mid x_1) = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1), \quad \mathrm{Var}(X_2 \mid x_1) = \sigma_2^2(1-\rho^2).$$

(3) X_1 and X_2 are independent if and only if $\rho = 0$.
(4) Any linear combination $aX_1 + bX_2$ has a normal distribution.

A remarkable property of the bivariate normal distribution is that zero covariance (uncorrelatedness) implies independence.

Corollary 2.1. *If (X_1, X_2) has a bivariate normal distribution and $\rho = 0$, then X_1 and X_2 are independent.*

7. Exercises

EXERCISE 2.1. *2.1.1 Let* $f(x_1, x_2) = 4x_1x_2$, $0 < x_1 < 1$, $0 < x_2 < 1$, *zero elsewhere, be the pdf of* X_1 *and* X_2 . *Find* $P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1)$, $P(X_1 = X_2)$, $P(X_1 < X_2)$, *and* $P(X_1 \leq X_2)$.

EXERCISE 2.2. *2.1.3 Let* $F(x, y)$ *be the distribution function of* X *and* Y . *For all real constants* $a < b$, $c < d$, *show that*

$$P(a < X \leq b, c < Y \leq d) = F(b, d) - F(b, c) - F(a, d) + F(a, c).$$

EXERCISE 2.3. *2.1.7 Let* $f(x, y) = e^{-x-y}$, $0 < x < \infty$, $0 < y < \infty$, *zero elsewhere, be the pdf of* X *and* Y . *Then if* $Z = X + Y$, *compute* $P(Z \leq 0)$, $P(Z \leq 6)$, *and, more generally,* $P(Z \leq z)$, *for* $0 < z < \infty$. *What is the pdf of* Z *?*

EXERCISE 2.4. *2.1.11 Let* X_1 *and* X_2 *have the joint pdf* $f(x_1, x_2) = 15x_1^2x_2$, $0 < x_1 < x_2 < 1$, *zero elsewhere. Find the marginal pdfs and compute* $P(X_1 + X_2 \leq 1)$.

EXERCISE 2.5. *2.2.5 Let* X_1 *and* X_2 *be continuous random variables with the joint probability density function* $f_{X_1,X_2}(x_1, x_2), -\infty < x_i < \infty, i = 1, 2$. *Let* $Y_1 = X_1 + X_2$ *and* $Y_2 = X_2$.

(1) Find the joint pdf f_{Y_1,Y_2} .

(2) Show that

$$f_{Y_1}(y_1) = \int_{-\infty}^{\infty} f_{X_1,X_2}(y_1 - y_2, y_2)\, dy_2,$$

which is sometimes called the convolution formula.

EXERCISE 2.6. *2.3.1 Let* X_1 *and* X_2 *have the joint pdf* $f(x_1, x_2) = x_1 + x_2$, $0 < x_1 < 1$, $0 < x_2 < 1$, *zero elsewhere. Find the conditional mean and variance of* X_2 , *given* $X_1 = x_1$, $0 < x_1 < 1$.

EXERCISE 2.7. *2.4.1 Let the joint pdf of* X_1 *and* X_2 *be*

$$f(x_1, x_2) = \begin{cases} 2 & 0 < x_1 < x_2 < 1 \\ 0 & \textit{elsewhere.} \end{cases}$$

Are X_1 *and* X_2 *independent? Why or why not?*

REMARK. ***Chapter 2 focuses on bivariate distributions: joint pmf/pdf, marginal distributions, conditional distributions, independence, transformations, and the correlation coefficient. Key formulas: (2.1.1) joint cdf, (2.1.8)/(2.1.9) marginal pmf/pdf, (2.1.10)/(2.1.11) expectation, (2.3.1)/(2.3.2) conditional pmf/pdf, (2.3.1)/(2.3.2) conditional expectation.***

CHAPTER 3

Some Special Distributions

1. The Binomial and Related Distributions

In Chapter 1 we introduced the uniform distribution and the hypergeometric distribution. In this chapter we discuss some other important distributions of random variables frequently used in statistics. We begin with the binomial and related distributions.

A Bernoulli experiment is a random experiment, the outcome of which can be classified in but one of two mutually exclusive and exhaustive ways, for instance, success or failure (e.g., female or male, life or death, nondefective or defective). A sequence of Bernoulli trials occurs when a Bernoulli experiment is performed several independent times so that the probability of success, say p , remains the same from trial to trial.

Let X be a random variable associated with a Bernoulli trial by defining it as follows:

$$X(\text{success}) = 1 \quad \text{and} \quad X(\text{failure}) = 0.$$

That is, the two outcomes, success and failure, are denoted by one and zero, respectively. The pmf of X can be written as

$$p(x) = p^x(1-p)^{1-x}, \quad x = 0, 1, \tag{18}$$

and we say that X has a Bernoulli distribution. The expected value of X is

$$\mu = E(X) = (0)(1-p) + (1)(p) = p,$$

and the variance of X is

$$\sigma^2 = \text{Var}(X) = p^2(1-p) + (1-p)^2 p = p(1-p).$$

It follows that the standard deviation of X is $\sigma = \sqrt{p(1-p)}$.

In a sequence of n independent Bernoulli trials, where the probability of success remains constant, let X_i denote the Bernoulli random variable associated with the ith trial. In such a sequence of Bernoulli trials, we are often interested in the total number of successes and not in the order of their occurrence. If we let the random variable X equal the number of observed successes in n Bernoulli trials, the possible values of X are $0, 1, 2, \ldots, n$. If x successes occur, where $x = 0, 1, 2, \ldots, n$, then $n - x$ failures occur. The number of ways of selecting the x positions for the x successes in the n trials is

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}.$$

Since the trials are independent and the probabilities of success and failure on each trial are, respectively, p and $1-p$, the probability of each of these ways is $p^x(1-p)^{n-x}$. Thus the pmf of X is

$$p(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x = 0, 1, 2, \ldots, n \\ 0 & \text{elsewhere.} \end{cases} \tag{19}$$

A random variable X that has a pmf of the form of $p(x)$ is said to have a binomial distribution, and any such $p(x)$ is called a binomial pmf. A binomial distribution is denoted by the symbol $b(n, p)$. The constants n and p are called the parameters of the binomial distribution.

Example 3.1. *Suppose we roll a fair six-sided die 3 times. What is the probability of getting exactly 2 sixes? For our notation, let X be the number of sixes obtained in the 3 rolls. Then X has a binomial distribution with $n = 3$ and $p = 1/6$. Hence,*

$$P(X = 2) = p(2) = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = 0.06944.$$

The mgf of a binomial distribution is easily obtained as follows:

$$M(t) = \sum_x e^{tx} p(x) = \sum_{x=0}^{n} \binom{n}{x} (pe^t)^x (1-p)^{n-x} = [(1-p) + pe^t]^n.$$

The mean μ and the variance σ^2 of X are

$$\mu = np, \quad \sigma^2 = np(1-p).$$

Theorem 3.1. *Suppose $X_1, X_2, \ldots, X_m$ are independent random variables such that X_i has binomial $b(n_i, p)$ distribution, for $i = 1, 2, \ldots, m$. Let $Y = \sum_{i=1}^m X_i$. Then Y has a binomial $b(\sum_{i=1}^m n_i, p)$ distribution.*

This result follows from the mgf or by convolution of pmfs.

1.1. Negative Binomial and Geometric Distributions. Consider a sequence of independent Bernoulli trials with constant probability p of success. Let the random variable Y denote the total number of failures in this sequence before the rth success, that is, $Y + r$ is equal to the number of trials necessary to produce exactly r successes with the last trial as a success. Here r is a fixed positive integer. The pmf of Y is

$$p_Y(y) = \begin{cases} \binom{y+r-1}{r-1} p^r (1-p)^y & y = 0, 1, 2, \ldots \\ 0 & \text{elsewhere.} \end{cases} \tag{20}$$

A distribution with a pmf of the form $p_Y(y)$ is called a negative binomial distribution. If $r = 1$, then Y has the pmf

$$p_Y(y) = p(1-p)^y, \quad y = 0, 1, 2, \ldots, \tag{21}$$

zero elsewhere. In this special case, $r = 1$, we say that Y has a geometric distribution. In terms of Bernoulli trials, Y is the number of failures until the first success.

The geometric distribution has the memoryless property:

$$P(Y \geq k + j \mid Y \geq k) = P(Y \geq j).$$

1.2. Multinomial Distribution. The binomial distribution is generalized to the multinomial distribution as follows. Let a random experiment be repeated n independent times. On each repetition, there is one and only one outcome from one of k categories. Call the categories $C_1, C_2, \ldots, C_k$. For each $i = 1, \ldots, k$, let p_i be the probability that the outcome is an element of C_i and assume that p_i remains constant throughout the n independent repetitions. Define the random variable X_i to be equal to the number of outcomes that are elements of C_i , $i = 1, 2, \ldots, k-1$. Because $X_k = n - X_1 - \cdots - X_{k-1}$, X_k is determined by the other X_i's.

The joint pmf of $(X_1, X_2, \ldots, X_{k-1})$ is

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_{k-1} = x_{k-1}) = \frac{n!}{x_1! \cdots x_{k-1}! x_k!} p_1^{x_1} \cdots p_{k-1}^{x_{k-1}} p_k^{x_k}, \tag{22}$$

for all $x_1, x_2, \ldots, x_{k-1}$ that are nonnegative integers and such that $x_1 + x_2 + \cdots + x_{k-1} \leq n$, where $x_k = n - x_1 - \cdots - x_{k-1}$ and $p_k = 1 - \sum_{j=1}^{k-1} p_j$.

We say that $(X_1, X_2, \ldots, X_{k-1})$ has a multinomial distribution with parameters n and $p_1, \ldots, p_{k-1}$.

1.3. Hypergeometric Distribution. Suppose we have a lot of N items of which D are defective. Let X denote the number of defective items in a sample of size n . If the sampling is done without replacement, the pmf of X is

$$p(x) = \frac{\binom{N-D}{n-x}\binom{D}{x}}{\binom{N}{n}}, \quad x = 0, 1, \ldots, n, \tag{23}$$

where, as usual, a binomial coefficient is taken to be 0 when the top value is less than the bottom value. We say that X has a hypergeometric distribution with parameters (N, D, n) .

The mean of X is

$$E(X) = n\frac{D}{N},$$

and the variance is

$$\mathrm{Var}(X) = n\frac{D}{N}\frac{N-D}{N}\frac{N-n}{N-1}. \tag{24}$$

The last term is often thought of as the correction term when sampling without replacement. Note that it is close to 1 if N is much larger than n .

2. The Poisson Distribution

Recall that the following series expansion holds for all real numbers z :

$$e^z = \sum_{x=0}^{\infty} \frac{z^x}{x!}.$$

Consider the function $p(x)$ defined by

$$p(x) = \begin{cases} \frac{\lambda^x e^{-\lambda}}{x!} & x = 0, 1, 2, \ldots \\ 0 & \text{elsewhere,} \end{cases} \tag{25}$$

where $\lambda > 0$. Since $\lambda > 0$, then $p(x) \geq 0$ and

$$\sum_{x=0}^{\infty} p(x) = \sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = 1;$$

that is, $p(x)$ satisfies the conditions of being a pmf of a discrete type of random variable. A random variable that has a pmf of the form $p(x)$ is said to have a Poisson distribution with parameter λ .

The Poisson distribution occurs in many areas of applications. Consider a process that counts the number of certain events occurring over an interval of time; for example, the number of tornados that touch down in Michigan per year, the number of cars entering a parking lot between 8:00 and 12:00 on a weekday, the number of car accidents at a busy intersection per week, the number of typographical errors per page of a manuscript, and the number of blemishes on a manufactured car door.

The mean and variance of a Poisson distribution are both λ :

$$E(X) = \lambda, \quad \mathrm{Var}(X) = \lambda.$$

The mgf of a Poisson distribution is

$$M(t) = \exp\{\lambda(e^t - 1)\}.$$

THEOREM 3.2. *If* $X_n \sim \mathrm{Bin}(n, p_n)$ *and* $np_n \to \lambda$ *as* $n \to \infty$ *, then* $X_n \xrightarrow{d} \mathrm{Poisson}(\lambda)$.

This theorem shows that the Poisson distribution can be used as an approximation to the binomial distribution when n is large and p is small.

3. The Γ, χ^2, and β Distributions

In this section we discuss several continuous distributions that are related to the gamma function.

3.1. The Gamma Distribution. The gamma function is defined by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx, \quad \alpha > 0.$$

The gamma function satisfies the recursion $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$, and for positive integers, $\Gamma(n) = (n-1)!$.

Definition 3.1 (Gamma Distribution). *A random variable X is said to have a gamma distribution with parameters $\alpha > 0$ and $\beta > 0$ if its pdf is*

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & x > 0 \\ 0 & \textit{elsewhere.} \end{cases}$$

We write $X \sim \Gamma(\alpha, \beta)$.

The parameters α and β are called the shape and scale parameters, respectively. The mean and variance of X are

$$E(X) = \alpha\beta, \quad \mathrm{Var}(X) = \alpha\beta^2.$$

The mgf of a gamma distribution is

$$M(t) = (1-\beta t)^{-\alpha}, \quad t < \frac{1}{\beta}.$$

Two important special cases of the gamma distribution are:

- $\Gamma(1, \beta)$ is the exponential distribution $\mathrm{Exp}(1/\beta)$.
- $\Gamma(\alpha, 2)$ with $\alpha = n/2$ is related to the chi-square distribution.

If $X_1 \sim \Gamma(\alpha_1, \beta)$ and $X_2 \sim \Gamma(\alpha_2, \beta)$ are independent, then $X_1 + X_2 \sim \Gamma(\alpha_1 + \alpha_2, \beta)$.

3.2. The χ^2-Distribution.

Definition 3.2 (Chi-Square Distribution). *Let $Z_1, Z_2, \ldots, Z_n$ be iid $N(0,1)$ random variables. Then the random variable*

$$X = \sum_{i=1}^n Z_i^2$$

is said to have a chi-square distribution with n degrees of freedom. We write $X \sim \chi^2(n)$.

The chi-square distribution is a special case of the gamma distribution with $\alpha = n/2$ and $\beta = 2$. That is, $\chi^2(n) \equiv \Gamma(n/2, 2)$.

The pdf of $X \sim \chi^2(n)$ is

$$f(x) = \begin{cases} \frac{1}{\Gamma(n/2)2^{n/2}} x^{n/2-1} e^{-x/2} & x > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

The mean and variance of a chi-square distribution are

$$E(X) = n, \quad \mathrm{Var}(X) = 2n.$$

The chi-square distribution has the additive property: if $X_1 \sim \chi^2(m)$ and $X_2 \sim \chi^2(n)$ are independent, then $X_1 + X_2 \sim \chi^2(m+n)$.

3.3. The β-Distribution.

Definition 3.3 (Beta Distribution). *A random variable X is said to have a beta distribution with parameters $\alpha > 0$ and $\beta > 0$ if its pdf is*

$$f(x) = \begin{cases} \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1} & 0 < x < 1 \\ 0 & \textit{elsewhere,} \end{cases}$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the beta function.

The mean and variance of $X \sim \mathrm{Beta}(\alpha, \beta)$ are

$$E(X) = \frac{\alpha}{\alpha+\beta}, \quad \mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Note that $\mathrm{Beta}(1, 1)$ is the uniform distribution on $(0, 1)$.

4. The Normal Distribution

The normal distribution is one of the most important distributions in statistics.

Definition 3.4 (Normal Distribution). *A random variable X is said to have a normal distribution with parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ if its pdf is*

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad -\infty < x < \infty.$$

We write $X \sim N(\mu, \sigma^2)$.

When $\mu = 0$ and $\sigma^2 = 1$, we say that X has a standard normal distribution, denoted by $Z \sim N(0, 1)$. The pdf of Z is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty,$$

and the cdf is $\Phi(z) = P(Z \leq z)$.

If $X \sim N(\mu, \sigma^2)$, then the standardized variable $Z = (X - \mu)/\sigma \sim N(0, 1)$. This property is extremely useful for computing probabilities:

$$P(X \leq x) = P\left(Z \leq \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

The mean and variance of $X \sim N(\mu, \sigma^2)$ are $E(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$.

The mgf of X is

$$M(t) = \exp\left\{\mu t + \frac{\sigma^2 t^2}{2}\right\}.$$

THEOREM 3.3 (Linear Combinations of Normals). *If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then for any constants a and b ,*

$$aX_1 + bX_2 \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2).$$

More generally, any linear combination of independent normal random variables is normally distributed.

5. The Multivariate Normal Distribution

5.1. Bivariate Normal Distribution. We say that (X_1, X_2) has a bivariate normal distribution if the joint pdf is given by

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right\},$$

for $-\infty < x_1 < \infty$, $-\infty < x_2 < \infty$, where $\mu_1 \in \mathbb{R}$, $\mu_2 \in \mathbb{R}$, $\sigma_1 > 0$, $\sigma_2 > 0$, and $-1 < \rho < 1$. We write $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$.

The parameters have the following interpretations:

- $E(X_1) = \mu_1$ and $E(X_2) = \mu_2$.
- $\mathrm{Var}(X_1) = \sigma_1^2$ and $\mathrm{Var}(X_2) = \sigma_2^2$.
- ρ is the correlation coefficient between X_1 and X_2 .

THEOREM 3.4 (Properties of Bivariate Normal). *If $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, then:*

(1) The marginal distributions are $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$.

(2) The conditional distribution of X_2 given $X_1 = x_1$ is normal with mean and variance:

$$E(X_2 \mid x_1) = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1), \quad \mathrm{Var}(X_2 \mid x_1) = \sigma_2^2(1-\rho^2).$$

(3) X_1 and X_2 are independent if and only if $\rho = 0$.

(4) Any linear combination $aX_1 + bX_2$ has a normal distribution.

6. t- and F-Distributions

6.1. The t-Distribution.

Definition 3.5 (t-Distribution). *Let $Z \sim N(0,1)$ and $X \sim \chi^2(n)$ be independent. Then the random variable*

$$T = \frac{Z}{\sqrt{X/n}}$$

is said to have a t-distribution with n degrees of freedom. We write $T \sim t(n)$.

The pdf of $T \sim t(n)$ is

$$f(t) = \frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\,\Gamma(n/2)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}, \quad -\infty < t < \infty.$$

The mean and variance of T are

$$E(T) = 0 \quad (n > 1), \quad \mathrm{Var}(T) = \frac{n}{n-2} \quad (n > 2).$$

As $n \to \infty$, $t(n) \xrightarrow{d} N(0,1)$.
When $n = 1$, the t-distribution is the Cauchy distribution.

6.2. The F-Distribution.

Definition 3.6 (F-Distribution). *Let $X_1 \sim \chi^2(m)$ and $X_2 \sim \chi^2(n)$ be independent. Then the random variable*

$$F = \frac{X_1/m}{X_2/n}$$

is said to have an F-distribution with degrees of freedom m and n . We write $F \sim F(m,n)$.

The pdf of $F \sim F(m,n)$ is

$$f(x) = \frac{\Gamma((m+n)/2)}{\Gamma(m/2)\Gamma(n/2)} \left(\frac{m}{n}\right)^{m/2} x^{m/2-1} \left(1 + \frac{mx}{n}\right)^{-(m+n)/2}, \quad x > 0.$$

The mean and variance of F are

$$E(F) = \frac{n}{n-2} \quad (n > 2), \quad \mathrm{Var}(F) = \frac{2n^2(m+n-2)}{m(n-2)^2(n-4)} \quad (n > 4).$$

Note that if $F \sim F(m,n)$, then $1/F \sim F(n,m)$.

Exercises

EXERCISE 3.1. *3.1.1 If the mgf of a random variable* X *is* $\left(\frac{1}{3}+\frac{2}{3}e^t\right)^5$ *, find* $P(X = 2 \text{ or } 3)$ *.*

EXERCISE 3.2. *3.1.3 If* X *is* $b(n,p)$ *, show that* $E(X/n) = p$ *and* $E[(X/n-p)^2] = p(1-p)/n$ *.*

EXERCISE 3.3. *3.1.11 Toss two nickels and three dimes at random. Make appropriate assumptions and compute the probability that there are more heads showing on the nickels than on the dimes.*

EXERCISE 3.4. *3.2.1 Let* X *have a Poisson distribution with parameter* $\lambda = 3$ *. Find* $P(X = 2)$ *and* $P(X \geq 1)$ *.*

EXERCISE 3.5. *3.3.1 Show that the gamma function* $\Gamma(\alpha)$ *satisfies* $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ *.*

EXERCISE 3.6. *3.3.3 Let* $X \sim \chi^2(10)$ *. Find* $P(X > 15.99)$ *using R.*

EXERCISE 3.7. *3.4.1 Let* $X \sim N(100, 225)$ *(i.e.,* $\sigma = 15$ *). Find:*

(1) $P(X < 85)$ *;*
(2) $P(90 < X < 115)$ *;*
(3) $P(X > 120)$ *.*

EXERCISE 3.8. *3.4.3 If* $X_1 \sim N(50, 36)$ *and* $X_2 \sim N(30, 16)$ *are independent, find the distribution of* $Y = 2X_1 - 3X_2$ *.*

EXERCISE 3.9. *3.5.1 If* (X, Y) *has a bivariate normal distribution with* $\mu_1 = 2$ *,* $\mu_2 = 3$ *,* $\sigma_1^2 = 4$ *,* $\sigma_2^2 = 9$ *, and* $\rho = 1/2$ *, find* $E(Y \mid X = 2)$ *and* $\mathrm{Var}(Y \mid X = 2)$ *.*

EXERCISE 3.10. *3.6.1 Let* $T \sim t(10)$ *. Find* $P(|T| > 2.228)$ *.*

EXERCISE 3.11. *3.6.5 Let* $F \sim F(5, 10)$ *. Find* $E(F)$ *(if it exists) and* $P(F > 3.33)$ *.*

REMARK. ***Chapter 3 covers special distributions: binomial, negative binomial, Poisson, gamma, chi-square, beta, normal, multivariate normal,*** t***, and*** F ***distributions. Key formulas: (3.1.2) binomial pmf, (3.2.1) Poisson pmf, (3.3.1) gamma pdf, (3.4.1) normal pdf. Important relationships: binomial*** $\rightarrow$ ***Poisson (large*** n***, small*** p***), chi-square is special gamma,*** t ***and*** F ***involve normal and chi-square.***

CHAPTER 4

Some Elementary Statistical Inferences

1. Sampling and Statistics

In Chapter 2, we introduced the concepts of samples and statistics. We continue with this development in this chapter while introducing the main tools of inference: confidence intervals and tests of hypotheses.

In a typical statistical problem, we have a random variable X of interest, but its pdf $f(x)$ or pmf $p(x)$ is not known. Our ignorance about $f(x)$ or $p(x)$ can roughly be classified in one of two ways:

(1) $f(x)$ or $p(x)$ is completely unknown.

(2) The form of $f(x)$ or $p(x)$ is known down to a parameter θ , where θ may be a vector.

For now, we consider the second classification, although some of our discussion pertains to the first classification also. Some examples are the following:

(a) X has an exponential distribution $\text{Exp}(\theta)$, where θ is unknown.

(b) X has a binomial distribution $b(n, p)$, where n is known but p is unknown.

(c) X has a gamma distribution $\Gamma(\alpha, \beta)$, where both α and β are unknown.

(d) X has a normal distribution $N(\mu, \sigma^2)$, where both the mean μ and the variance σ^2 of X are unknown.

We often denote this problem by saying that the random variable X has a density or mass function of the form $f(x; \theta)$ or $p(x; \theta)$, where $\theta \in \Omega$ for a specified set Ω . For example, in (a) above, $\Omega = \{\theta \mid \theta > 0\}$. We call θ a parameter of the distribution. Because θ is unknown, we want to estimate it.

Suppose, in particular, we have a random variable X that has a distribution dependent on a parameter θ . Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from this distribution; that is, the X_i are iid (independent and identically distributed). A statistic is a function $T = g(X_1, X_2, \ldots, X_n)$ of the sample that does not depend on any unknown parameters. In this chapter we discuss several statistics that are used to estimate the unknown parameter θ or to test hypotheses about θ .

Let $X_1, X_2, \ldots, X_n$ denote a random sample of size n from a distribution that has mean $\mu = E(X_i)$ and variance $\sigma^2 = \text{Var}(X_i)$. The sample mean is defined by

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i. \tag{26}$$

The sample mean has the following properties:

$$E(\bar{X}) = \mu, \quad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

By the Weak Law of Large Numbers, $\bar{X}$ converges in probability to μ as $n \to \infty$. By the Central Limit Theorem, for large n ,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

The sample variance is defined by

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2. \tag{27}$$

An unbiased estimator of σ^2 is S^2 because $E(S^2) = \sigma^2$.

Other important statistics include the sample standard deviation $S = \sqrt{S^2}$, the sample median, and the sample standard deviation.

1.1. Point Estimators. A point estimator $\hat{\theta}$ of a parameter θ is a statistic that provides a single value as an estimate of θ .

Several criteria are used to evaluate estimators:

Definition 4.1 (Bias). *The bias of an estimator $\hat{\theta}$ is defined as $E(\hat{\theta}) - \theta$. If $E(\hat{\theta}) = \theta$, then $\hat{\theta}$ is unbiased.*

Definition 4.2 (Mean Squared Error). *The mean squared error of an estimator $\hat{\theta}$ is* $\mathrm{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$. *For unbiased estimators,* $\mathrm{MSE} = \mathrm{Var}(\hat{\theta})$.

Example 4.1. *Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2 . Then $\bar{X}$ is an unbiased estimator of μ because $E(\bar{X}) = \mu$. The variance of $\bar{X}$ is σ^2/n . The mean squared error of $\bar{X}$ is σ^2/n .*

Theorem 4.1 (Method of Moment Matching). *The kth sample moment is $M_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k$. The method of moments equates sample moments to population moments to obtain estimators.*

Example 4.2. *Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$. The maximum likelihood estimators (MLEs) are*

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

Note that $\hat{\mu}$ is unbiased, but $\hat{\sigma}^2$ is biased (though asymptotically unbiased).

2. Confidence Intervals

A confidence interval provides a range of plausible values for an unknown parameter.

DEFINITION 4.3 (Confidence Interval). *Let θ be an unknown parameter. Let $L = g_1(X_1, \ldots, X_n)$ and $U = g_2(X_1, \ldots, X_n)$ be two statistics such that $L \leq U$. If for all $\theta \in \Omega$,*

$$P_\theta(L \leq \theta \leq U) = 1 - \alpha,$$

then the random interval $[L, U]$ is called a $100(1-\alpha)\%$ confidence interval for θ .

The quantity $1-\alpha$ is called the confidence coefficient, and α is the significance level.

EXAMPLE 4.3 (Confidence interval for normal mean, known variance). *Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$ where σ^2 is known. Then a $100(1-\alpha)\%$ confidence interval for μ is*

$$\left[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right],$$

where $z_{\alpha/2}$ satisfies $\Phi(z_{\alpha/2}) = 1 - \alpha/2$.

EXAMPLE 4.4 (Confidence interval for normal mean, unknown variance). *Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$ where both μ and σ^2 are unknown. Then a $100(1-\alpha)\%$ confidence interval for μ is*

$$\left[\bar{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}, \quad \bar{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right],$$

where $t_{\alpha/2,n-1}$ is the upper $\alpha/2$ critical value of the $t(n-1)$ distribution.

THEOREM 4.2 (Confidence interval for variance). *Let $X_1, X_2, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$. A $100(1-\alpha)\%$ confidence interval for σ^2 is*

$$\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \quad \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right],$$

where $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ are critical values of the $\chi^2(n-1)$ distribution.

2.1. Confidence Intervals for Difference in Means. Let $X_1, \ldots, X_{n_1}$ be a random sample from $N(\mu_1, \sigma_1^2)$ and $Y_1, \ldots, Y_{n_2}$ be an independent random sample from $N(\mu_2, \sigma_2^2)$.

If σ_1^2 and σ_2^2 are known, a $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\left[\bar{X} - \bar{Y} \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right].$$

If $\sigma_1^2 = \sigma_2^2 = \sigma^2$ but unknown, the pooled estimator is

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}.$$

A $100(1-\alpha)\%$ confidence interval is

$$\left[\bar{X}-\bar{Y} \pm t_{\alpha/2, n_1+n_2-2} S_p \sqrt{\frac{1}{n_1}+\frac{1}{n_2}}\right].$$

2.2. Confidence Interval for Difference in Proportions. Let $X \sim \operatorname{Bin}(n_1, p_1)$ and $Y \sim \operatorname{Bin}(n_2, p_2)$ be independent. Large-sample approximations yield the following $100(1-\alpha)\%$ confidence interval for $p_1 - p_2$:

$$\left[\hat{p}_1-\hat{p}_2 \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}\right],$$

where $\hat{p}_1 = X/n_1$ and $\hat{p}_2 = Y/n_2$.

3. Order Statistics

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with cdf $F(x)$. Define the order statistics by arranging the sample in ascending order:

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}.$$

The kth order statistic is $X_{(k)}$. The sample median is $X_{((n+1)/2)}$ if n is odd, or the average of the two middle values if n is even.

The sample range is $R = X_{(n)} - X_{(1)}$ and the sample interquartile range is $IQR = X_{(3n/4)} - X_{(n/4)}$ (approximately).

The cdf and pdf of the kth order statistic $X_{(k)}$ are:

$$F_{X_{(k)}}(x) = \sum_{j=k}^{n} \binom{n}{j} [F(x)]^j [1-F(x)]^{n-j},$$

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1-F(x)]^{n-k} f(x).$$

3.1. Quantiles. The population quantile x_p is defined by $F(x_p) = p$.

Definition 4.4 (Sample Quantile). *The sample quantile x_p can be estimated by the order statistic $X_{(\lceil np \rceil)}$ or by linear interpolation.*

Common quantiles include:

- Median: $p = 0.5$
- Quartiles: $p = 0.25, 0.75$
- Deciles: $p = 0.1, 0.2, \ldots, 0.9$
- Percentiles: $p = 0.01, 0.02, \ldots, 0.99$

4. Introduction to Hypothesis Testing

A statistical hypothesis is a statement about the distribution of a random variable. We test a null hypothesis H_0 against an alternative hypothesis H_1 .

DEFINITION 4.5 (Null and Alternative Hypotheses). *The null hypothesis H_0 is the hypothesis being tested. The alternative hypothesis H_1 is the hypothesis we accept if H_0 is rejected.*

The four possible outcomes of a hypothesis test are:

	Fail to Reject H_0	Reject H_0
H_0 True	Correct	Type I Error (α)
H_0 False	Type II Error (β)	Correct Power ($1-\beta$)

The significance level $\alpha = P(\text{Type I Error}) = P(\text{Reject } H_0 \mid H_0 \text{ true})$.

EXAMPLE 4.5 (Test for normal mean, known variance). *Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with known σ^2 . To test $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$, we use the test statistic*

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

We reject H_0 at level α if $Z > z_\alpha$.

EXAMPLE 4.6 (Test for normal mean, unknown variance). *Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with unknown σ^2 . To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, we use the test statistic*

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

We reject H_0 at level α if $|T| > t_{\alpha/2,n-1}$.

5. Chi-Square Tests

The chi-square test is used for categorical data.

5.1. Goodness-of-Fit Test. Let $X_1, \ldots, X_n$ be iid observations from a categorical distribution with k categories. Under H_0, the probabilities are $p_{1,0}, p_{2,0}, \ldots, p_{k,0}$. The observed counts are $O_1, \ldots, O_k$ and the expected counts are $E_i = np_{i,0}$.

The chi-square goodness-of-fit statistic is

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}.$$

Under H_0 , this statistic has approximately a $\chi^2(k-1)$ distribution for large n . We reject H_0 if $\chi^2 > \chi^2_{\alpha,k-1}$.

5.2. Test of Independence. For a contingency table with r rows and c columns, let O_{ij} be the observed count in cell (i, j) . The expected count under independence is $E_{ij} = (O_{i\cdot}O_{\cdot j})/n$.

The chi-square test statistic for independence is

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Under H_0 of independence, $\chi^2 \approx \chi^2((r-1)(c-1))$.

6. The Method of Monte Carlo

Monte Carlo methods use random sampling to approximate statistical quantities.

The law of large numbers states that if $E[g(X)] = \mu$ exists, then

$$\frac{1}{n}\sum_{i=1}^{n} g(X_i) \xrightarrow{P} \mu \quad \text{as } n \to \infty.$$

Thus, we can approximate μ by simulating n values from the distribution of X and averaging $g(X_i)$.

Example 4.7 (Approximating π). *We can approximate π by simulating points in the unit square $[0, 1] \times [0, 1]$ and computing the proportion that fall inside the unit quarter circle. The area of the quarter circle is $\pi/4$, so $\hat{\pi} = 4 \times$ (proportion inside) .*

6.1. Accept-Reject Generation Algorithm. The accept-reject algorithm generates random samples from a target distribution $f(x)$ using a proposal distribution $g(x)$ with known density.

Example 4.8 (Accept-Reject Algorithm). *(1) Choose a proposal density $g(x)$ such that $f(x) \le Mg(x)$ for all x and some constant $M > 1$.*

(2) Generate $Y \sim g$ and $U \sim$ Uniform$(0, 1)$ independently.

(3) If $U \le f(Y)/(Mg(Y))$, accept $X = Y$; otherwise, reject and return to step 2.

The acceptance probability is $1/M$. The algorithm generates samples from f exactly.

7. Bootstrap Procedures

Bootstrap methods use resampling to estimate the sampling distribution of a statistic.

Let $X_1, \ldots, X_n$ be a random sample and $T = g(X_1, \ldots, X_n)$ be a statistic. The bootstrap proceeds as follows:

(1) Resample n observations with replacement from the original sample to form a bootstrap sample $X_1^*, \ldots, X_n^*$.

(2) Compute the bootstrap replicate $T^* = g(X_1^*, \ldots, X_n^*)$.

(3) Repeat steps 1-2 many times (say B times) to obtain $T_1^*, \ldots, T_B^*$.
(4) Approximate the distribution of T by the empirical distribution of $T_1^*, \ldots, T_B^*$.

7.1. Percentile Bootstrap Confidence Intervals. A $100(1-\alpha)\%$ percentile bootstrap confidence interval for θ is

$$\left[T^*_{(\alpha/2)}, T^*_{(1-\alpha/2)}\right],$$

where $T^*_{(k)}$ is the kth order statistic of the bootstrap replicates.

Exercises

Exercise 4.1. *4.1.1 Let X_1, X_2, X_3, X_4, X_5 be a random sample of size 5 from a distribution with mean $\mu = 10$ and variance $\sigma^2 = 4$. Find the mean and variance of $\bar{X}$.*

Exercise 4.2. *4.1.3 Let $X_1, \ldots, X_n$ be iid with mean μ and variance σ^2 . Show that S^2 is an unbiased estimator of σ^2 .*

Exercise 4.3. *4.2.1 Let $X_1, \ldots, X_{16}$ be a random sample from $N(50, 25)$. Find $P(48 < \bar{X} < 52)$.*

Exercise 4.4. *4.2.3 Construct a 95% confidence interval for the mean of a normal distribution based on the following sample: $n = 25$, $\bar{x} = 100$, $s = 15$.*

Exercise 4.5. *4.3.1 Two independent samples are given: Sample 1: $n_1 = 30$, $\bar{x}_1 = 75$, $s_1 = 10$; Sample 2: $n_2 = 40$, $\bar{x}_2 = 80$, $s_2 = 12$. Assuming equal variances, find a 90% confidence interval for $\mu_1 - \mu_2$.*

Exercise 4.6. *4.4.1 Let $X_1, \ldots, X_{10}$ be a random sample from $N(0, 1)$. Find the probability that the sample median exceeds 0.5.*

Exercise 4.7. *4.5.1 A manufacturer claims that the mean lifetime of a light bulb is at least 1000 hours. A sample of 25 bulbs has a mean lifetime of 950 hours with a standard deviation of 80 hours. Test the claim at the $\alpha = 0.05$ level.*

Exercise 4.8. *4.7.1 A chi-square goodness-of-fit test is performed on a die to test whether it is fair. The die is rolled 120 times with the following frequencies: Face 1: 18, Face 2: 22, Face 3: 16, Face 4: 24, Face 5: 19, Face 6: 21. Compute the test statistic and p-value.*

Exercise 4.9. *4.8.1 Describe the accept-reject algorithm for generating random samples from a gamma distribution using an exponential proposal distribution.*

Exercise 4.10. *4.9.1 Explain the bootstrap procedure for estimating the standard error of the sample median.*

Remark. ***Chapter 4 covers statistical inference: sampling, point estimation, confidence intervals, hypothesis testing, chi-square tests, Monte Carlo methods, and bootstrap procedures. Key concepts: sample mean and variance, unbiased estimation, CI formulas for normal means and variances, hypothesis testing framework, Type I/II errors, bootstrap resampling.***

1. 问答记录

问题记录. 问题: ... **回答:** ...

学习笔记.